

Supplementary Material:

Exact Symmetry-Stratified Search for Fast Matrix Multiplication over $\mathbb{F}_2$

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1 The two rank-47 schemes

Both schemes below are Γ -invariant with orbit type $[12, 6, 6, 6, 6, 3, 3, 3, 1, 1]$, satisfy all 4096 Brent equations by direct evaluation, and share no term. Machine-readable copies are in `alphatensor_444_rank47.json` and `records/gamma47_B_rank47.json`, both under `results/`. The script `experiments/paper_tables.py` regenerates the table below from those files and re-verifies both schemes before printing.

1.1 Reading the bitmasks

Each factor is a 4×4 matrix over $\mathbb{F}_2$ encoded as an integer: bit $4i + j$ (least significant bit first) holds entry (i, j) . For example, $a_t = 1$ is the matrix with a single 1 in position $(0, 0)$, and $a_t = 4096$ is the matrix with a single 1 in position $(3, 0)$. Term t contributes the rank-one tensor $a_t \otimes b_t \otimes c_t$ under the index convention of Equation (1) of the main paper, with the a factor indexed (i, j) , the b factor (j, k) and the c factor (k, i) .

1.2 Invariants of the two schemes

	AlphaTensor	This work
Rank	47	47
Host-verified over $\mathbb{F}_2$	yes	yes
Invariant under Γ ($ \Gamma = 12$)	yes	yes
Orbit sizes	12, 6, 6, 6, 6, 3, 3, 3, 1, 1	12, 6, 6, 6, 6, 3, 3, 3, 1, 1
Stabilizer orders	1, 2, 2, 2, 2, 4, 4, 4, 12, 12	1, 2, 2, 2, 2, 4, 4, 4, 12, 12
a -factor ranks	{1:20, 2:12, 3:15}	{1:19, 2:12, 3:16}
b -factor ranks	{1:20, 2:12, 3:15}	{1:19, 2:12, 3:16}
c -factor ranks	{1:9, 2:24, 3:13, 4:1}	{1:13, 2:24, 3:9, 4:1}
Shared factors ($a/b/c$)	0 / 0 / 0	0 / 0 / 0
Lifts to $\mathbb{Z}/4$	no	no
Terms in common		0
Related by an element of Ω ($ \Omega = 82,944$)		no

The weight histograms (total number of set bits across the three factors of a term) also differ: {5:3, 7:3, 8:6, 9:13, 10:9, 12:12, 15:1} for AlphaTensor’s scheme, against {10:6, 12:3, 14:6, 16:12, 17:12, 22:1, 24:3, 25:1, 27:3} for the new one. The second scheme is markedly denser: its terms carry roughly twice as many nonzero entries.

2 Encoding details

2.1 Instance construction

A census or rewrite instance is fixed by a triple: a group $G \leq \text{GL}(4, 2)^3 \rtimes S_3$, a multiset of subgroups $H_1, \dots, H_m \leq G$ (the *slot list*) with $\sum_i [G : H_i]$ equal to the target rank, and, for a rewrite instance, a residual tensor obtained by subtracting the kept orbits from the target. The encoder emits:

- $3n^2$ Boolean variables per slot for the representative term’s factor bits, constrained so that the representative is fixed by its slot’s subgroup;

Table 1: The two inequivalent $\mathbb{F}_2$ schemes for $\langle 4, 4, 4 \rangle$ of rank 47, term by term. Each factor is a 4×4 matrix over $\mathbb{F}_2$ written as a decimal bitmask: bit $4i + j$ is the (i, j) entry. Left: the AlphaTensor factorization as imported and re-verified. Right: the scheme found from scratch in this work. Both satisfy all 4096 Brent equations; they share no term.

AlphaTensor, rank 47				This work, rank 47			
t	a_t	b_t	c_t	t	a_t	b_t	c_t
1	1	1	261	1	11	4113	22796
2	2	16	4105	2	13	4353	854
3	4	256	19	3	14	4368	12442
4	8	4096	31711	4	94	5141	43267
5	16	37900	1538	5	109	6289	22019
6	32	22720	40962	6	112	49480	3
7	68	3477	2	7	158	37144	12291
8	84	1029	258	8	173	20801	771
9	100	2192	4098	9	176	7317	42499
10	136	53320	42498	10	221	22737	1539
11	152	36872	30210	11	238	37917	40963
12	168	20544	43778	12	781	4504	12389
13	257	923	4	13	877	34824	29729
14	512	12456	49156	14	925	39312	12321
15	769	152	4100	15	1291	4882	49301
16	871	144	4096	16	1792	43041	5
17	920	8	21504	17	2573	12577	85
18	1024	39426	100	18	2827	39698	49157
19	1281	770	20	19	3083	38929	20485
20	2056	47136	49252	20	3181	37017	21505
21	2568	12320	49364	21	3218	2184	4097
22	3080	38912	28772	22	3341	12728	101
23	3176	2048	12320	23	3584	5018	49253
24	3224	36864	29728	24	12302	4421	937
25	4096	12358	3080	25	12382	17472	35617
26	4368	14	1024	26	12462	21765	801
27	4744	12360	33792	27	24587	4897	3161
28	8194	885	8	28	25867	8226	34001
29	8736	224	32768	29	27147	13059	1105
30	12290	69	264	30	28672	25618	9
31	12375	5	256	31	30583	4095	1
32	12456	64	35072	32	31717	27629	32769
33	16384	22048	168	33	36878	12562	153
34	17472	3584	32	34	38155	13104	32913
35	18472	22560	32800	35	39428	546	17
36	24578	800	24	36	45067	22305	3081
37	25863	768	16	37	47062	42974	1025
38	27144	32	32912	38	48059	61695	33793
39	32776	29698	3240	39	49163	21521	2313
40	33816	37890	1056	40	49246	20565	35073
41	34944	57344	33824	41	49313	1092	257
42	36872	12290	3288	42	53248	4965	3241
43	38152	2	1104	43	56797	65295	1057
44	39432	12288	34000	44	57008	60522	33825
45	49160	21504	2984	45	57358	12615	169
46	49240	1024	800	46	60787	52919	33
47	49320	20480	35616	47	61166	65520	32801

- Tseitin variables for each distinct degree-3 monomial $a[i, j] b[j, k] c[k, i]$ appearing anywhere, defined by the usual AND clauses;
- one XOR constraint per G -orbit of tensor coordinates, expanded into CNF, equating the sum of monomials over all terms with the residual value at a representative coordinate;
- nonzero clauses forbidding an all-zero factor on any axis of any slot, which is sound because a term with a zero factor contributes nothing and can be deleted, lowering the rank.

2.2 The two exact reductions

E1 (orbit equations). An encoded scheme is a sum of complete G -orbits of terms, so the value of the encoded tensor at coordinates P and gP is the same expression for every $g \in G$. Emitting one equation per orbit of coordinates is therefore an exact rewriting, not a relaxation. For Γ at $n = 4$ this replaces the 4096 equations by 379.

E2 (gate sharing). The same degree-3 monomials recur across the surviving equations. Introducing one Tseitin variable per distinct monomial, cached across the whole instance, removes the duplication.

Validation. Both reductions were validated by (i) recovering AlphaTensor’s 47 from its own orbit type under the reduced encoding, with the decoded scheme identical to the input and host-verified, and (ii) an exhaustive $n = 2$ cross-check against the unreduced encoder: 44 of 44 census instances agreed on verdict. Measured effect on a from-scratch *4imes4* census instance of an order-12 group with orbit type $[12, 12, 12, 6, 3, 1]$:

	egincenter	egintabularllrrr	oprul	E1	E2	Variables	Clauses	Equations						
off	off	561,440	1,872,962	4096	off	on	210,720	820,802	4096	on	off	52,896	176,578	384
on	on	34,304	120,802	384	ottomrule									
oindent That is a 16.4-fold reduction in variables and a 15.5-fold reduction in clauses. Measured effect on solve time, on the instances that motivated the change:														

Instance	Old encoder, CaDiCaL 1.5.3	E1+E2, CaDiCaL 1.9.5
census group 7	191 s	1.6 s
census group 33	139 s	25 s
census group 2	undecided	19 s
hardest orbit-rewrite shape	10^8 conflicts	≤ 29 s

2.3 Blocking solutions

Same-rank searches must exclude the input scheme, and enumeration of distinct schemes must exclude what has already been found. Blocking the satisfying *assignment* is insufficient: the same term set reappears under a different assignment of representatives to slots. We block at the *set* level, adding for each blocked scheme a clause asserting that at least one term differs, using equality gadgets over the decoded term bits. An earlier version of this campaign blocked at the assignment level, which caused two over-counts that were found and corrected before publication: a set of same-rank 4×4 instances that had returned only relabelings had not in fact been decided (closed afterwards by 237 UNSAT instances), and a set of 15 result files at 5×5 was found to contain 6 distinct schemes rather than 15.

3 The fixed-point filter in practice

3.1 Reach

Of the 64 cyclic conjugacy classes of the ambient permutation group Ω at $n = 4$:

Classes with an involution	59
of which the involution has fixed coordinates	19
with 64 fixed coordinates ($\langle 2, 2, 2 \rangle$ fixed format)	10
with 256 fixed coordinates ($\langle 2, 2, 4 \rangle$)	6
with 1024 fixed coordinates ($\langle 2, 4, 4 \rangle$)	3
Classes of odd order (no involution)	5

For the 40 classes whose involution has no fixed coordinate the projected system is empty and the filter has no power; this is visible as the flat row at zero in Figure 5 of the main paper.

3.2 Power

Scope	Multisets	Count bound	SAT filter	Filter budget	Survivors*
All 64 classes	11,767	416	476	465	10,875
The 19 reachable classes	4,236	416	476	465	3,344
Group 6 (order 12, $\langle 2, 2, 4 \rangle$)	300	27	148	5	120
Groups 20, 21 (order 6, $\langle 2, 4, 4 \rangle$)	600	238	22	329	11

*Survivors include the instances on which the filter itself exhausted its 50,000-conflict budget, which are passed on rather than refuted. The whole pass took 14 minutes on 10 processes and produced 892 exact refutations. Its own hard cases are the near-tight small-format instances (14–22 fixed terms for a rank-14 target), where CDCL needs minutes; a 50,000-conflict cap with the policy “budget exhausted means survive” was adopted, which keeps the filter sound by construction.

3.3 Small-format ranks

The corollary needs the bilinear rank of the fixed format. Computed by SAT over $\mathbb{F}_2$:

Format	Result	Evidence
$\langle 2, 2, 2 \rangle$	rank = 7	rank ≤ 6 UNSAT in 35.9s; rank ≤ 7 SAT, verified
$\langle 2, 2, 4 \rangle$	rank ≤ 14	rank ≤ 13 undecided at 5×10^6 conflicts
$\langle 2, 4, 4 \rangle$	rank ≥ 16	flattening bound; not tightened

For a swap-type involution the restricted terms are rank-one matrices and the bound is the matrix rank of the restricted target, which is 4 at $n = 4$.

4 Full campaign tallies

Every campaign below is an exhaustive enumeration over the stated instance family, with one JSON record per instance in `results/*.jsonl`. “Solver hours” is the sum of the recorded per-instance solve times, so it excludes encoding and process overhead.

Campaign	UNSAT	SAT	Undecided	Solver hours
4×4 rank 46, ≤ 2 orbits deleted, refills ≤ 4	4,011	0	0	9.0
4×4 rank 46, ≤ 2 orbits, refills 5–17	7,729	0	0	15.1
4×4 rank 46, 3 orbits deleted, refills ≤ 4	5,765	0	0	17.7
4×4 same rank, ≤ 2 orbits	1,578	54	0	8.6
4×4 same rank, 3 orbits	5,669	51	17	35.7
closure of the 17 open shapes (all class combinations)	237	0	0	0.7
4×4 rank 46 around the second 47	1,763	0	0	0.07
4×4 same rank around the second 47	1,961	0	0	0.07
4×4 rank 46 from scratch (cyclic census, 3 h)	7	0	93	28.9
5×5 rank 92 from the record	648	0	0	0.2
5×5 rank 92, 3 orbits, from the record	6,851	0	0	15.9
5×5 rank 92 over 64 family members	41,472	0	0	8.9
5×5 rank 92, 3 orbits, two far seeds	9,635	0	0	17.9
5×5 same rank from the record	681	435	0	1.0
6×6 rank 152 from the record	1,578	0	0	1.7
6×6 rank 152 over 11 family members	16,311	0	0	20.2

The 51 and 54 SAT outcomes in the 4×4 same-rank rows all decoded to relabelings of the input scheme and were blocked; after the closure pass, no Γ -invariant rank-47 scheme other than the input exists in that neighbourhood.

4.1 Solve-time distributions

Campaign group	Outcome	n	Median (s)	p90 (s)	Max (s)
4×4 orbit rewrites	UNSAT	20,714	3.4	20.5	11,369.5
	SAT	595	26.8	151.9	989.4
	budget	22	1186.1	1498.2	1903.6
5×5 / 6×6 rewrites	UNSAT	3,555	1.9	5.1	8.3
	SAT	2,544	2.4	4.5	10.0
4×4 census, from scratch	UNSAT	7	137.0	170.4	191.0
	budget	93	1001.0	1919.7	3173.1

The consistent pattern — refutations in seconds to minutes, budget exhaustions two to three orders of magnitude slower — is the empirical basis for the budget policy: short sweep budgets, a wall-time watchdog per instance, and targeted re-runs at a hundred times the budget only for recorded exhaustions.

5 Census state at the freeze

The two censuses that were still running when the machine was needed elsewhere are enumerated, checkpointed per instance and resumable. Their state is frozen in the file `investigation_state.json` under `results/`.

Census	Processed	Exact refutations	Undecided / surviving
Γ -invariant rank 46 (filter stage)	23,072 of 47,210	21,647	1,425 surviving
Odd-order $GL(4, 2)^3$ cyclic	1,861 of 4,321	1,795	66 undecided at 300 s
Diagonal permutation subgroups	452	441	11 undecided

The odd-order census is complete for groups of order 15 (1,668 UNSAT, 28 undecided) and order 21 (127 UNSAT, 38 undecided); orders 35 and 105 are not started. The diagonal permutation census refuted the first 150 orbit-type multisets of $S_4 \times C_3$ (100 by the count bound, 11 by the filter, 39 by full solves) and of $S_4 \times S_3$ (120 by the count bound, 30 by the filter), and left 9 undecided in the diagonal S_4 family.

6 The rank-49 attractor

All 3,257 host-verified rank-49 schemes found by three different flip dynamics have the factor-rank signature $\{1:36, 2:12, 4:1\}$ on each of the three axes — identical to Strassen^{⊗2}'s. Since the signature is a $\text{GL}(4, 2)^3$ -invariant, they all lie in the same equivalence family, and no walk left it.

Two steering strategies were tested and retired:

- **Forbid rank-4 factors.** The gated walk did leave the attractor (floor 50, 2,278 distinct signatures), but 5,883 of 6,000 rank-50 ancestors of the 47 — generated by plus transitions from the 47 itself — contain a rank-4 factor, and the gated pool shares zero signatures with the ancestor set. The gate excludes the target's basin by construction.
- **Stay in the τ -quotient.** From a trivial start the quotient walk plateaus at rank 55 with 13 τ -fixed terms and 21 pairs, whereas every symmetric rank-55 ancestor of the 47 has 11 fixed terms; the two sets of 1,594 and 1,371 signatures share nothing.

A useful by-product of the second attempt: in the τ -quotient, without a move that changes a fixed term's a -factor, the fixed-term count is frozen and only even ranks are reachable. The identity $(a + e)(a + e)^\top = aa^\top + ae^\top + ea^\top + ee^\top$ yields four symmetric moves, two of which are $3 \rightarrow 2$ reductions the plain flip graph cannot express. They lifted τ -symmetric 3×3 walks from 25 to 23 and freed the parity at 4×4 , changing the floor from 56 to 55.

7 Process findings

These are engineering observations, recorded because they cost real time to learn and are not specific to this problem.

Finding	Rule adopted
Conflict budgets do not bound wall time: instances with a 10^6 -conflict budget ran 15–53 minutes when the conflict rate collapsed.	A wall-time watchdog per instance, independent of the conflict budget.
Decidable instances decide fast; budgets are consumed by undecidable ones.	Short sweep budgets, then targeted re-runs at 10^8 conflicts only for recorded exhaustions.
Conjugacy deduplication of stabilizers is exact and large: $9,605 \rightarrow 1,763$ instances in one sweep, $87,371 \rightarrow 5,737$ in another.	Always enumerate over conjugacy classes, never over subgroups.
Folding kept structure into a residual: $574,000 \rightarrow 4,400$ variables.	Never encode what is already decided.
Assumption-based incremental SAT was slower than fresh instances (822s versus 3.5s on the same instance).	Fresh instances, with the encoding template cached.
Lex-leader symmetry breaking is sound but direction-dependent: about $2\times$ faster on UNSAT, $10\text{--}64\times$ slower on satisfiable controls.	Opt-in, for refutation sweeps only.
12 physical cores, not 24: 24 processes cut throughput about sixfold.	At most 12–16 solver workers on small instances, 6–8 on large ones.
Static striding creates stragglers: 2 of 12 workers held 249 instances.	A cost-sorted work queue with a per-instance wall cap.
Two PySAT bindings misbehaved: Kissat 4.0.4 returned a wrong answer on a five-clause formula and then crashed; CaDiCaL 3.0.0 crashed silently.	Both excluded; all results use CaDiCaL 1.9.5.
Process-kill verification by name count missed live processes twice.	List full command lines before declaring a run stopped.

8 Reproducibility

8.1 What is archived

- **results/records/** — the imported and newly found record-rank schemes, including `gamma47_B_rank47.json` (the second rank-47 scheme).
- **results/*.json** — every host-verified scheme produced, each carrying its `scheme_bitmasks` and a `verified_brent.f2` flag.
- **results/*.jsonl** — one record per SAT instance for every campaign, carrying the instance key, the outcome and the solve time.
- **results/investigation.state.json** — frozen tallies of the two censuses still in progress.
- **results/certificates/** — DRAT proofs emitted through Glucose 4.2 for a sample of theorem-grade refutations.

The public repository carries all of the above, and the Zenodo record archives that repository exactly, so the two hold the same files. Two categories of intermediate data are deliberately omitted from both: the solver stdout and stderr files, which are process output rather than results, and the bulk flip-graph walker pools, whose contents the dissection digests summarise.

8.2 Regenerating this paper’s content

From `experiments/`:

Command	Effect
<code>python reproduce.py</code>	21 checks of the headline counts against <code>results/</code>
<code>python paper_data.py</code>	recomputes every number in both papers into <code>paper_data.json</code>
<code>python paper_figures.py</code>	regenerates <code>figures/*.pdf</code> and <code>*.png</code>
<code>python paper_tables.py</code>	regenerates <code>tables_main.tex</code> , <code>tables_schemes.tex</code>
<code>python fp_filter.py --selftest</code>	validates the fixed-point lemma, including the $n = 2$ cross-check
<code>python encode_opt.py --selftest</code>	validates the E1+E2 encoding against the unreduced encoder
<code>python glsym.py --selftest</code>	validates the general-linear symmetry encoder

`paper_data.py` re-verifies every scheme it reports against the Brent equations and re-aggregates every campaign tally from the per-instance logs, so the numbers in the papers are not transcribed from any narrative document.

8.3 System requirements

Python 3.9 or later, `python-sat` (for CaDiCaL 1.9.5), NumPy and Matplotlib. The GPU walkers additionally need CuPy and a CUDA device; they are not required to reproduce any result in the papers, all of which are CPU SAT computations or direct verifications.